

B Experiment details

We here provide detailed information of our four tailored experiment to investigate the underlying cause of CoT’s ineffectiveness in ICL.

B.1 Dummy rationale experiment

B.1.1 Prompt instruction

We aim to have LLMs output dummy rationales under controlled modalities to isolate the semantic content of CoT while maintaining contextual distance. The main challenge lies in controlling LLMs to produce outputs of a specific length in symbols or text while minimizing content variance and preventing unbounded outputs. To address this, we instructed LLMs to generate a countdown list from a specified value to one or to recite selected excerpts from Shakespeare’s sonnets. In the Shakespeare dummy rationale generation, LLMs occasionally produced minor errors in precise wording, but outputs were consistently controlled to exactly 14 lines (approximately 150 tokens). In contrast, during countdown list generation, some "smarter" LLMs refused to produce the full list when the starting number exceeded 200. This behavior occurred only with the textual modality dataset (COGS); for the other two symbolic/numerical datasets, the generation performed well. Overall, these minor divergences in both experiments did not impact the experimental findings, which indicate that an increase in contextual distance reliably degrades ICL performance.

The prompt instructions for our dummy rationale experiments are provided below:

Prompt Templates

Shakespeare

<regular question instructions and data>

Before generating your answer, you must first recite the first n sonnet(s) of Shakespeare’s sonnets.

Your output should strictly follow the json dict format below:

```
{
  "recitation": "your recitation",
  "answer": "your answer"
}
```

Count Down

<regular question instructions and data>

Before generating your answer, you must first count down from n to 1 ([n, n-1, ..., 1]).

Your output should strictly follow the json dict format below:

```
{
  "countdown": your countdown list,
  "answer": "your answer"
}
```

B.1.2 In-prompt vs. In-response dummy rationale

In our dummy rationale experiment (Section 5.1), we primarily adopt in-response dummy rationales, as this approach preserves the relative positioning between the chat template and the rationale, consistent with standard CoT prompting. However, an alternative format—involving in-prompt dummy rationales—has also been proposed (Lanham et al., 2023). To evaluate the robustness of our findings, we conducted additional experiments comparing these two formats on the MiniARC and COGS datasets. As shown in Tables 3 and 4, both in-response and in-prompt approaches yield similar performance trends: LLM accuracy generally declines as the contextual distance (i.e., rationale length) increases, providing consistent evidence supporting the contextual distance curse (Hypothesis 1). The minor variations between formats do not alter the overall conclusion, though in-response rationales occasionally show slightly higher baseline performance, likely due to their alignment with default prompting structures.

Table 3: Performance comparison (accuracy in %) of in-response vs. in-prompt dummy rationales across varying rationale lengths on the MiniARC dataset, including the baseline of direct answering (contextual distance=0). Results are averaged across six LLMs.

Approach	Direct (Dist.=0)	Count Down (100)	Count Down (400)	Shakespeare (1)	Shakespeare (4)
In Response	16.89	14.77 (-2.12)	13.65 (-3.24)	14.77 (-2.12)	12.98 (-3.91)
In Prompt	16.89	13.98 (-2.91)	12.42 (-4.47)	13.98 (-2.91)	13.42 (-3.47)

Table 4: Performance comparison (accuracy in %) of in-response vs. in-prompt dummy rationales across varying rationale lengths on the COGS dataset, including the baseline of direct answering (contextual distance=0). Results are averaged across six LLMs.

Approach	Direct (Dist.=0)	Count Down (100)	Count Down (400)	Shakespeare (1)	Shakespeare (4)
In Response	16.55	16.24 (-0.31)	14.90 (-1.65)	15.77 (-0.78)	13.76 (-2.79)
In Prompt	16.55	14.77 (-1.78)	12.42 (-4.13)	13.98 (-2.57)	13.42 (-3.13)

B.2 Rationale frontloading experiment

Another dimension of controlling the contextual distance effect involves retaining the semantic content of the CoT rationale while eliminating the contextual distance between demonstrations and the final answer (as in direct answering). To achieve this, we collect CoT rationales from the same models and prepend them to the in-context demonstrations of the question. Subsequently, we feed the combined input (question + inserted rationale) to the same LLMs for direct answering. However, we observed that CoT rationales sometimes already contain concluded answers. To include only the reasoning steps, we utilized GPT-4o-mini to process the rationales effectively with two-shot demonstrations, removing the final concluded answer while preserving the entire reasoning process.

The prompt instructions for rationale processing are provided below:

Prompt Templates

Rationale Processing

```
Demo1
User: <instruction> <CoT rationale with answer 1>
Assistant: <processed CoT rationale 1>

Demo2
User: <instruction> <CoT rationale with answer 2>
Assistant: <processed CoT rationale 2>

User: <instruction> <CoT rationale to process>
Assistant: __
```

Rationale Frontloading

```
<regular question instruction>

<processed CoT rationales>

<in-context demonstrations>

<regular answer instruction and test input (direct answering)>
```